CMS · NANOAOD · TRIGGER OBJECTS

The unseeded diphoton leg never gets its NanoAOD filter bits

Photon **TrigObj_filterBits** 16 and 17 (14 and 15 before NanoAOD v15) are documented for HLT_Diphoton30_22_R9Id_OR_IsoCaloId_AND_HE_R9Id_Mass90 / Mass95, but no photon trigger object ever carries them. Why it happens, how widespread it is, and the one-line fix.

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Checked against cms-sw/cmssw master ed35654b1 (today), release branches 13_0_X to 16_1_X, and the 2018 to 2026 HLT menus

`PhysicsTools/NanoAOD/python/triggerObjects_cff.py`

Outline

1 **The problem** — slides 3-6

the diphoton path and its two legs, the data path from HLT to NanoAOD, where the unseeded object is dropped, the photon bit list

2 **Scope and measurement** — slides 7-12

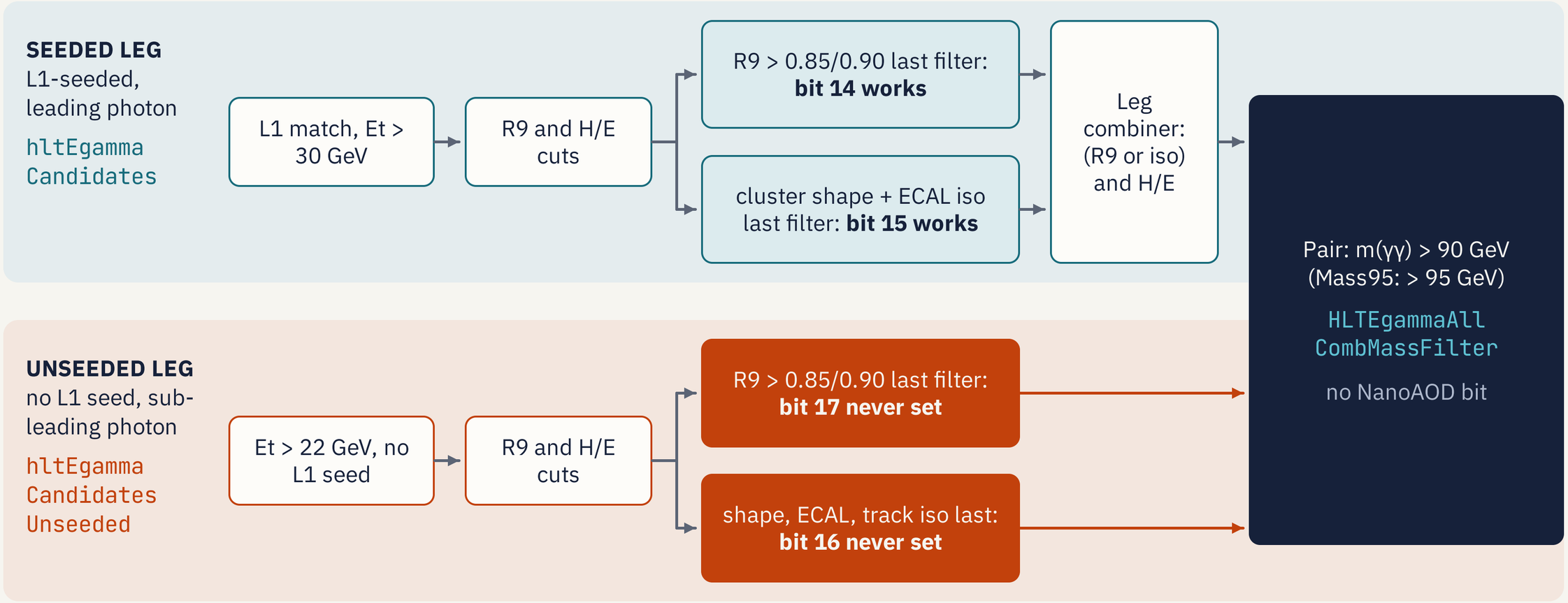
NanoAOD versions and HLT menus, the scan of the whole trigger-object table, thirteen files measured, the fix test, trigger efficiency for the Diphoton30_22 and Diphoton30_18 unseeded legs

3 **The fix and what follows** — slides 13-14

the one-line fix, validation, the pull request and the follow-ups it does not cover

One path, two legs, four last filters

HLT_Diphoton30_22_R9Id_OR_IsoCaloId_AND_HE_R9Id_Mass90 and Mass95 (same sequence), every menu 2018 to 2026



Orange boxes: the unseeded last filters NanoAOD discards. Bits in v15 numbering; v12 to v14: 12, 13, 15, 14.

From the HLT filter to TrigObj_filterBits: the object survives four hand-offs and dies at the fifth

`1 · HLT filters (saveTags)`	The unseeded last filters tag their objects with the collection **hltEgammaCandidatesUnseeded**; the leg's first filter saves them with type 92, the later ones with type 81.
`2 · hltTriggerSummaryAOD`	Objects are stored once per HLT collection; every filter that accepted an object records its key and the type it saved it with.
`3 · patTrigger → slimmedPatTrigger (MiniAOD)`	The pat object keeps the collection label **hltEgammaCandidatesUnseeded::HLT**, all its types (92 and 81) and the labels of every filter it passed.
`4 · unpackedPatTrigger (NanoAOD)`	Filter labels are unpacked; the object reaches the table producer intact, with its collection name and types.
`5 · triggerObjectTable, Photon selection`	**sel = type(92) && pt > 15 && coll('hltEgammaCandidates')** is evaluated before the quality bits. The collection test fails, the object is dropped, and bits 16 and 17 are never computed.

Where it breaks: the producer's decision for one unseeded photon object

1 **type(92)?** yes. The leg's first filter (HLT1Photon, triggerType 92) saved the object as TriggerCluster; PAT keeps every type.

2 **pt > 15?** yes. The unseeded leg requires Et > 22 GeV (18 GeV for the Diphoton30_18 paths).

3 **coll('hltEgammaCandidates')?** no. The stored label is hltEgammaCandidatesUnseeded::HLT. Without a wildcard the comparison is exact (TriggerObjectStandAlone::hasCollection falls back to TriggerObject::hasCollection).

4 **Discarded before the bits.** qualityBits is only evaluated for objects that pass sel; the duplicate merging only sees accepted objects. Bits 16 and 17 stay 0 in every event.

```
# triggerObjects_cff.py, Photon
sel = "type(92) && pt > 15 &&
       coll('hltEgammaCandidates')"
skipObjectsNotPassingQualityBits = True
```

```
// TriggerObjectTableProducer.cc, produce()
for (const auto &obj : trigObjs)
  for (const auto &sel : sels_)
    if (sel.match(obj)) { // fails here
      bits = sel.qualityBits(obj);
      if (bits =≠ 0) selected.push_back(...);
    }
```

Same logic in 13_0_X, 13_3_X, 14_0_X, 15_0_X and master; only the bit width (32 bits up to 15_0_1, 64 bits from 15_0_2) and the merging rule changed.

The photon bit list: what is documented versus what can be set

bit v15	bit v12 to v14	filter named by the bit	objects come from	status
8	8	`hltHtEcal800`	hltHtMhtEcal (HT object, type 89)	**never set**: not an e/gamma object, since v10
14	12	`hltEG30LR9Id85b90eHE12R9Id50b80eR9IdLastFilter`	hltEgammaCandidates	**works** (seeded leg)
15	13	`hltEG30LIso60CaloId15b35eHE12R9Id50b80e EcalIsoLastFilter`	hltEgammaCandidates	**works** (seeded leg)
16	14	`hltEG22Iso60CaloId15b35eHE12R9Id50b80e TrackIsoUnseededLastFilter`	hltEgammaCandidatesUnseeded	**never set**: collection rejected by sel
17	15	`hltEG22R9Id85b90eHE12R9Id50b80eR9UnseededLastFilter`	hltEgammaCandidatesUnseeded	**never set**: collection rejected by sel
18	16	`hltEG30Iso60CaloId15b35eR9Id50b90e HE12b10eR9Id50b80eEcalIsoFilter`	hltEgammaCandidates	**works for 2018 and 2022 menus**; filter renamed in 2023, stale since
19	17	`hltEG18TrackIso60Iso60CaloId15b35eR9Id50b90e HE12b10eR9Id50b80eTrackIsoUnseededFilter`	hltEgammaCandidatesUnseeded	**never set** (collection), and renamed in 2023
20	none	`hltEG*L1VBFLooseIsoEGHEFilter`	hltEgammaCandidates	**works** (filter exists from 2023)

Bits 0 to 7 (single-photon HE filters), 9 to 13 (EB tight-ID filters and the Mu17_Photon30 photon leg) are fine. Decimal values of the dead unseeded bits: 65536, 131072, 524288 in v15 and master; 16384, 32768, 131072 in v12 to v14.

How widespread (1): every NanoAOD version since v12, every menu since 2018, still there today

NanoAOD	release checked	photon selection	dead unseeded bits	affected
v9	CMSSW_10_6_X	seeded only, no photon bits at all	none defined	no photon TrigObj
v10, v11	CMSSW_12_4_X, 12_6_X	seeded only, pt > 20	bits absent	no
v12	CMSSW_13_0_0 to 13_0_17 (prod 13_0_13)	seeded only	14, 15, 17	yes
v13	CMSSW_13_3_1_patch1	seeded only	14, 15, 17	yes
v14	CMSSW_14_0_X, 14_1_X, 14_2_X	seeded only	14, 15, 17	yes
v15	CMSSW_15_0_2 to 15_0_15_patch4, 15_1_X	seeded only	16, 17, 19	yes
next	16_0_X, 16_1_X (tags to 16_1_3), master ed35654b1 today	seeded only, no open PR or issue	16, 17, 19	yes

Menus

19 menu files checked, 2018 GRun to the frozen 2026 menu: the four Diphoton30_22 filters exist everywhere, always on the same collections, with the same 16-filter chain. Path versions v13 (2018, early 2022) to v29 (2026).

Datasets

Every Run 3 NanoAOD file: 2022 to 2025 data (PromptNano, 22Sep2023, 16Dec2023, v15 re-NANO) and the Summer22, 22EE, 23, 23BPix and 24 MC campaigns, plus any Run 2 input re-processed with v12 or later.

Analyses

Any NanoAOD-based matching of the sub-leading photon finds nothing: HiggsDNA-style workflows, the HYgammaRunIII legend that lists bits 16 and 17 as the legs, FLAF configurations. MiniAOD-based flashgg reads the objects directly and is unaffected.

How widespread (2): every filter bit of every object type, resolved against six menus

object	bits	never settable (collection or type)	stale name in the 2025 menu	fine in 2025
Electron	20	none	2: WPLoose (gone 2023), triple-electron (gone 2025)	18
Photon	21	4: bits 16, 17, 19 (unseeded) and 8 (hltHtEcal800)	3: bit 12 (gone 2022), 18 and 19 (renamed 2023)	15
Muon	14	none	2: mu-tau DeepTau overlap filters (gone 2025)	12
Tau	31	none	1: DeepTau (gone 2025)	30
BoostedTau	3	v12 to v14: whole selection, see below	1: 2022 ParticleNetTauTau name	2
Jet	46	none	10: pixel-only HH4b filters (gone 2024), DeepCSV and DeepJet (gone 2025), 3-jet pt40 and pt45	34 (+2 L1 seeds)
FatJet	7	none	3: 2022 and 2023 soft-drop and ParticleNet names	4
HT, MHT	6, 2	none	2: Run 2 L1 seed names (never in Run 3 menus)	3 (+1), 2
MET	0	no bits with the skip flag on: MET objects never written	not applicable	none, all versions

Same mechanism elsewhere

Only in the Photon list: the three unseeded diphoton bits and bit 8, whose filter saves HT objects. Electron and Muon have no collection mismatch. Two whole tables were empty by configuration: MET in every version since v9, and BoostedTau in v12 to v14, whose selection required a filter name ending in ParticleNetTauTau while the menus end in 0p30 or Tag0p03 (relaxed in 15_0_2).

Stale names are the common case

One bit list serves the 2022 to 2026 menus, so bits for filters that were renamed or retired stay documented but silently empty for later years, and bits added for new filters are empty for earlier years. That is a different, documentation-level problem; the photon one is a code defect that no menu can cure.

Measured on real files: not one photon object with an unseeded bit, in any version since v12

NanoAOD	file (one per dataset, read on lxplus)	events	fired Mass90	photons: seeded bits	unseeded bits	e unseeded	id 2	id 1515
v9	UL2018 data, EGamma 2018D	300000	31771	no photon TrigObj	none defined	n/a	0	0
v10	2022 prompt data, EGamma 2022D	300000	46707	bits not defined	none defined	n/a	0	0
v11	2022 prompt data, EGamma 2022F	300000	41546	bits not defined	none defined	n/a	0	0
v12	Summer22 MC, GluGluHtoGG M125	12979	7725	15626 / 17978	0 / 0 / 0	n/a	0	0
v12	2022 data, EGamma 2022D 22Sep2023	300000	46407	208527 / 255099	0 / 0 / 0	n/a	0	0
v13	Summer23 MC, GluGluHto2G M125	17760	10453	20865 / 24170	0 / 0 / 0	702	0	0
v13	2022 data, EGamma 2022D 16Dec2023	300000	46794	207844 / 252774	0 / 0 / 0	29042	0	0
v14	2024 prompt data, EGamma0 2024D	300000	35892	203620 / 252039	0 / 0 / 0	19444	0	0
v15	Summer24 MC, GluGluH to 2G M125	4180	2406	4815 / 5588	0 / 0 / 0	205	0	22
v15	2022 data, EGamma 2022D NanoAODv15	300000	48687	212270 / 258546	0 / 0 / 0	30451	0	430
v15	2024 data, EGamma0 2024D MINIv6NANOv15	300000	30945	201628 / 248974	0 / 0 / 0	22980	0	254
133X	Winter24 MC, GluGluHToGG M125 powheg	175654	101598	205090 / 239250	0 / 0 / 0	9860	0	0
16_0_X	2026 data, EGamma0 2026C PromptReco	300000	1511	124972 / 145161	0 / 0 / 0	1034	0	42

Seeded bits are the two leading-leg last filters (12 and 13 up to v14, 14 and 15 from v15); unseeded bits are the three sub-leading-leg filters (14, 15, 17 up to v14; 16, 17, 19 from v15). Every prediction holds: unseeded bits never set, seeded bits present in every diphoton-triggered event, the electron unseeded bit populated from v13, no MET objects (id 2) in any version, no BoostedTau objects (id 1515) before v15. Measurement of 4 November 2026, last three rows 5 November; results/REPORT.md on the notes branch.

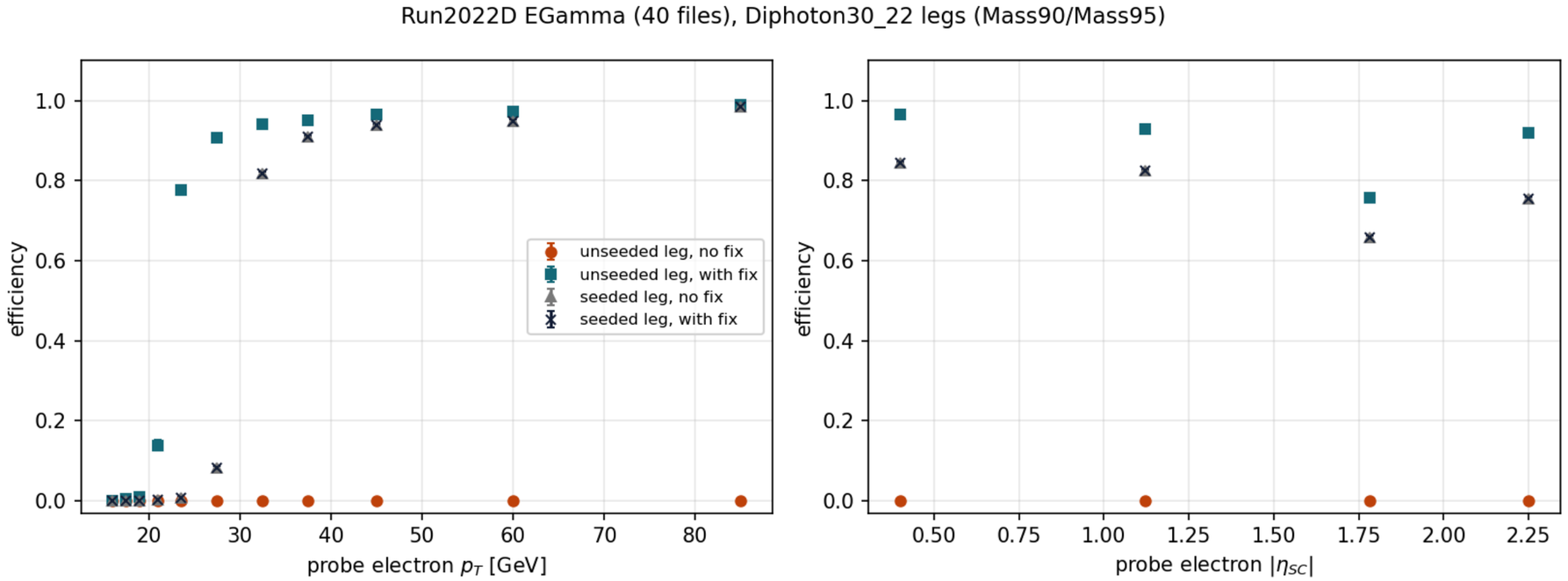
Fix test on real input: every Mass90 event gets its sub-leading-leg bits, nothing else changes

NANO file (v13, v15: 17760 events; 16_0_X: 20000)	fired Mass90	photons: seeded bits	photons: unseeded bits	Mass90 events with an unseeded bit	nTrigObj mean	size (bytes)
central v13 file	10453	20865 / 24170	0 / 0 / 0	0	5.287	29333713
v13 re-made, no fix	10453	20865 / 24170	0 / 0 / 0	0	5.287	29339920
v13 re-made, with fix	10453	20865 / 24170	20461 / 18365 / 0	10453 (all)	5.388	29363499
v15 re-made, no fix	10453	20865 / 24170	0 / 0 / 0	0	38.134	39947152
v15 re-made, with fix	10453	20865 / 24170	20461 / 18365 / 0	10453 (all)	38.225	39959608
16_0_X 2026C data, no fix	308	8909 / 10760	0 / 0 / 0	0	2.100	26190407
16_0_X 2026C data, with fix	308	8909 / 10760	671 / 1010 / 0	308 (all)	2.118	26206694

- Re-made without the fix, the file is identical to the central v13 file: same TrigObj (id, bits, pt) in 17760 of 17760 events.
- The 21254 photon objects that gain an unseeded bit have pt from 22.04 GeV (median 58.30 GeV), none below the 22 GeV leg threshold.
- Every object type other than photons is unchanged in 17760 of 17760 events; photon objects 24835 to 26639 (v13), 25188 to 26807 (v15).
- The third unseeded bit (Diphoton30_18 leg) stays 0: that filter name does not exist in the 2023 menu; also 0 in 2026 data (not investigated).
- 2026 data, CMSSW_16_0_4_patch2, 20000 events of a Run2026C MINIAOD file: unseeded bits in all 308 Mass90 events, none before; pt from 22.01 GeV (median 37.06 GeV). Not compared event by event with a central file.
- Relval in CMSSW_16_1_3 with the fix (12452.0, 141.046): builds and runs without exceptions; outputs identical to the clean release except 25 DQM TimerService histograms per workflow. Its NANO TrigObj table is empty in both areas, so it does not exercise the photon selection.

Seeded / unseeded bits: 12, 13 / 14, 15, 17 in v13; 14, 15 / 16, 17, 19 in v15. Input: the MiniAOD parent of the central Run3Summer23 v13 GluGluHto2G M125 file; CMSSW_13_3_1_patch1 (v13) and CMSSW_15_0_15_patch4 (v15). 5 November 2026, results/REPORT.md "Fix test" and "Relval in CMSSW_16_1_3".

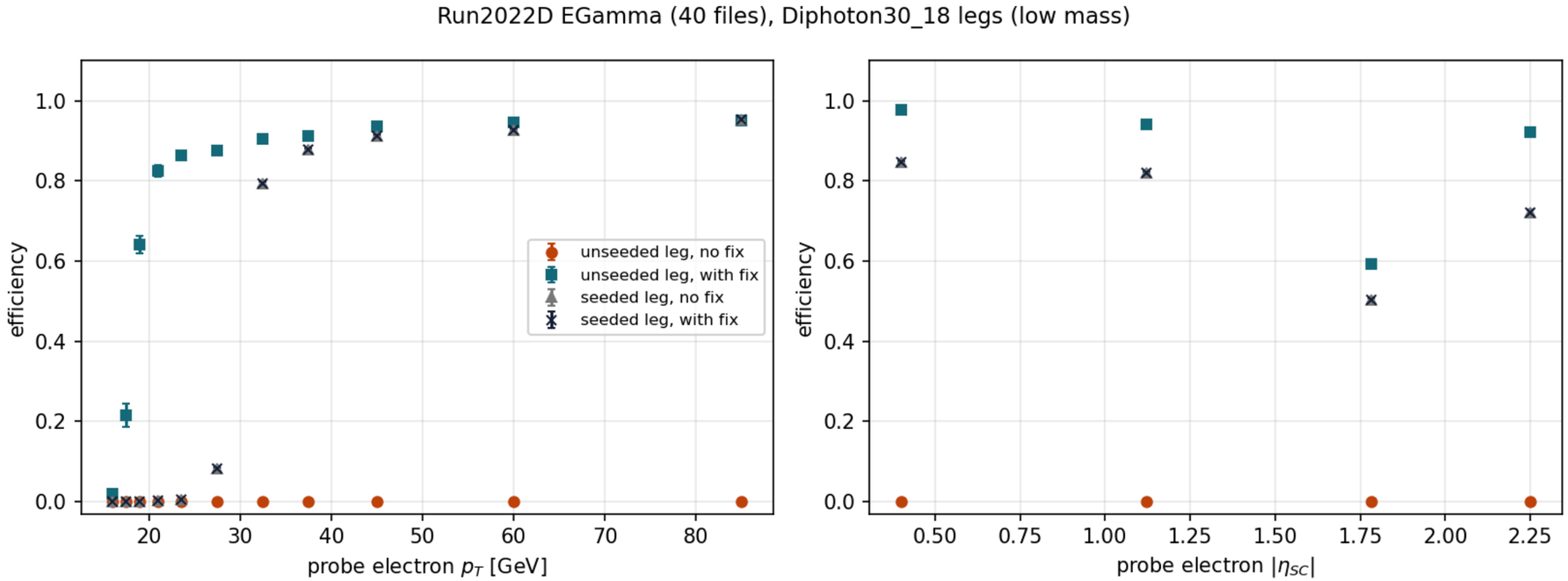
Trigger efficiency with the fix, Diphoton30_22: unseeded leg 0 to 0.92, seeded leg unchanged



	no fix	with fix
unseeded leg (bit 14 or 15)	0 / 59863	55133 / 59863 = 0.921 (0.920-0.922)
seeded leg (bit 12 or 13)	48108 / 59863 = 0.804	48108 / 59863 = 0.804, identical in every bin

Z to ee tag-and-probe on NANO re-made in CMSSW_13_3_1_patch1 from the 40 2022D EGamma 22Sep2023 MINIAOD parents of one central 16Dec2023 NANO file (3125939 events, HTCondor); without the fix the re-made TrigObj equals the central file in 19539 / 19539 shared events. Tag: tight electron, pT > 35 GeV, |eta_SC| < 2.1, matched to the HLT_Ele32_WPTight_Gsf object and to a seeded-leg photon object. Probe: loose electron, pT > 15 GeV, opposite charge, 70 < m_ee < 110 GeV (596 same-sign pairs). 68% Clopper-Pearson intervals. 2026C data (16_0_4_patch2, 20000 events): 42 pairs, unseeded leg 0 / 42 without and 39 / 42 with the fix.

Trigger efficiency with the fix, Diphoton30_18 (low mass): unseeded leg 0 to 0.91



	no fix	with fix
unseeded leg (EG18 TrackIso, bit 17)	0 / 57727	52460 / 57727 = 0.909 (0.908-0.910)
seeded leg (EG30 EcalIso, bit 16)	44989 / 57727 = 0.779	44989 / 57727 = 0.779, identical in every bin

Same 40 files, tag-and-probe and probe selection as the Diphoton30_22 slide; the tag matches the 30_18 seeded-leg bit instead (537 same-sign pairs). The unseeded-leg turn-on is at 18-20 GeV here and at 22-25 GeV for 30_22. The 1.566-2.0 |eta_SC| bin is lower for both legs; not investigated.

The fix: one line, mirroring what the Electron selection has done since 2023

```
PhysicsTools/NanoAOD/python/triggerObjects_cff.py, Photon = cms.PSet( ... )
- sel = cms.string("type(92) && pt > 15 && coll('hltEgammaCandidates')"),
+ sel = cms.string("type(92) && pt > 15 && (coll('hltEgammaCandidates') || coll('hltEgammaCandidatesUnseeded'))"),
```

Why this form

It names exactly the two e/gamma collections and copies the Electron selection (PR 41827). A wildcard coll('hltEgammaCandidates*') would also match, but would silently admit any future collection with that prefix.

What changes in the output

Unseeded objects enter the photon branch. With skipObjectsNotPassingQualityBits on, only those that set a photon bit are written, and the only photon-bit filters on the unseeded collection are the three diphoton ones. A seeded twin in the same direction is merged with OR-ed bits; otherwise a second photon entry appears, as for electrons.

What does not change

Bit numbers, documentation strings, the other object types, the event content. Existing files cannot be repaired: the objects were never written. Only a new NANO pass with the fix restores the sub-leading leg.

Status. Commit 1699a39f7 on the fork branch claude/fervent-lamport-3oiuh6 and the same change as aebf3eb64 on top of upstream master ed35654b1 (branch nano-photon-trigobj-unseeded). Applies cleanly to today's master. Both branches are on the fork santanumahata/cmssw; nothing pushed to cms-sw/cmssw, no pull request opened yet.

Validation, pull request, and the follow-ups this fix does not cover

Validation

- Done 4-5 November: thirteen NanoAOD files, v9 to v15 and 2026 data (16_0_X), data and MC; unseeded photon bits 0 in every file.
- Done 5 November: NANO re-made with the fix (13_3_1_patch1, 15_0_15_patch4): unseeded bits in all 10453 Mass90 events, nothing else changed; 2026 data in 16_0_4_patch2: all 308 Mass90 events.
- Done 5 November: relval 12452.0 and 141.046 in CMSSW_16_1_3: no exceptions, outputs identical except DQM timing; relval NANO has no TrigObj.

Pull request

- Branch nano-photon-trigobj-unseeded (commit aebf3eb64) is on the fork. Still to do: open it against cms-sw:master citing PR 40711 and PR 41827; xpog signature.
- Request a CMSSW_15_0_X backport if another NanoAOD production from that branch is planned.

Follow-ups, separate from the one-line fix

- Add the 2023 names of the Diphoton30_18 filters as alternatives inside bits 18 and 19, so the numbering stays stable.
- Move or drop bit 8 (hltHtEcal800): its objects are HT objects and can never be photons.
- Review the stale bits per year (WPLoose, triple-electron, DeepTau overlaps, pixel-only jet filters, DeepCSV and DeepJet) and say in the documentation which menus each bit covers.
- Bits for the low-mass eta1p5, DiphotonMVA14p25, Diphoton15_1x EBEB and DoublePhoton paths, with the EGM POG and the Higgs PAG.
- Inform XPOG that MET trigger objects are never written and that the BoostedTau table was empty in v12 to v14.

Notes, results, patch, PR text: santanumahata/cmssw, branch nanoaod-trigobj-notes. Nothing pushed to cms-sw/cmssw.